Heaven's light is our guide

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering

Sessional (Lab Report)-04

Course Name : Data Communication Sessional
Course Code : CSE 3204
Experiment No : 04
Experiment Name : Analysis of different types of line coding
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1.Experiment name:

Analysis of different types of line coding.

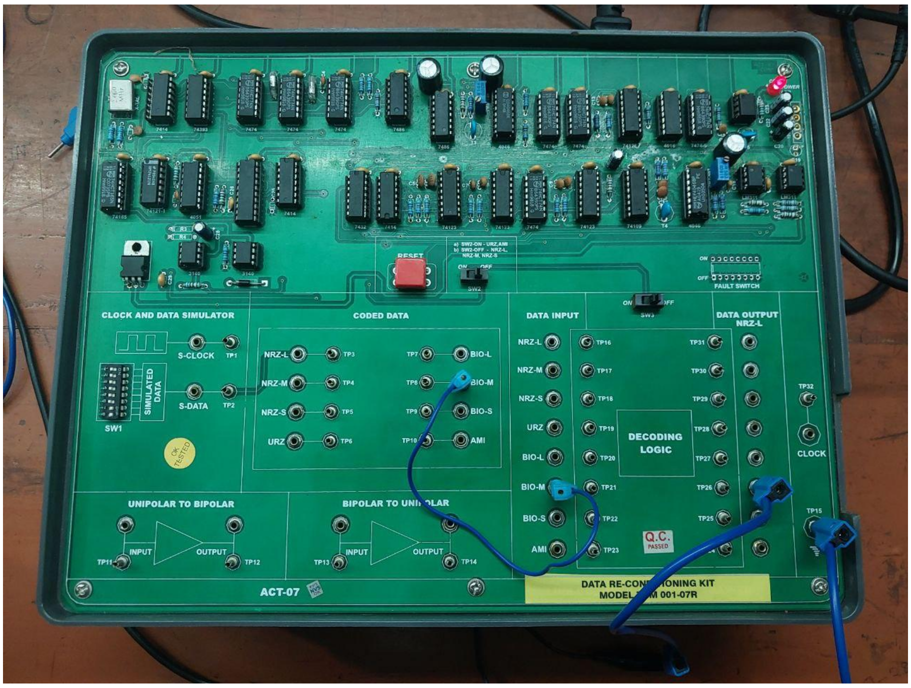
Objective:

- The primary objective of this study is to investigate the mechanics of various line coding techniques, specifically NRZ-L, NRZ-S, NRZ-M, and URZ-AMI.
- By implementing both encoding and decoding processes, the experiment helps to evaluate how these distinct schemes influence signal representation and transmission characteristics.

Theory:

Line coding is a fundamental mechanism in digital communication, responsible for translating binary streams into physical signals appropriate for channel transmission. This study examines four specific encoding methodologies:

- **NRZ-L:** Assigns distinct voltage levels to represent logical '1' and '0'.
- **NRZ-S:** Utilizes differential encoding, where a voltage transition signifies a '0' and a constant voltage represents a '1'.
- **NRZ-M:** The inverse of NRZ-S, where a transition indicates a '1' and the signal remains steady for a '0'.
- **URZ-AMI:** Maps logical '0' to zero voltage, while logical '1' alternates between positive and negative polarities. These schemes are evaluated based on their distinct characteristics regarding spectral efficiency, power consumption, and synchronization capabilities.



Required Equipment:

- Function Generator: To generate digital input signals.
- Oscilloscope: To visualize and analyze the encoded waveforms.
- Line Coding Circuit/Software: To implement different coding schemes.
- Computer with Signal Processing Software: For waveform analysis and comparison.

Experimental Setup:

The experimental setup consists of a function generator for creating input binary signals, a DSP Kit for encoding them into different line coding formats, and an oscilloscope for waveform visualization. A decoding mechanism is implemented to retrieve the original binary signal.

Procedure:

Encoding Process:

1. A binary input sequence was generated (e.g., 10110011).
    - A higher-frequency sine wave (~100–150 kHz typically but appears similar here due to time base settings) is used as the carrier.
2. The sequence was encoded using different line coding schemes (NRZ-L, NRZ-S, NRZ-M, URZ-AMI).
3. The encoded signals were observed using an oscilloscope.

Decoding Process:

1. The encoded signals were fed into the decoder.
2. The decoder analyzed the received waveform and reconstructed the original binary sequence.
3. The decoded sequence was verified against the original input.

Output:

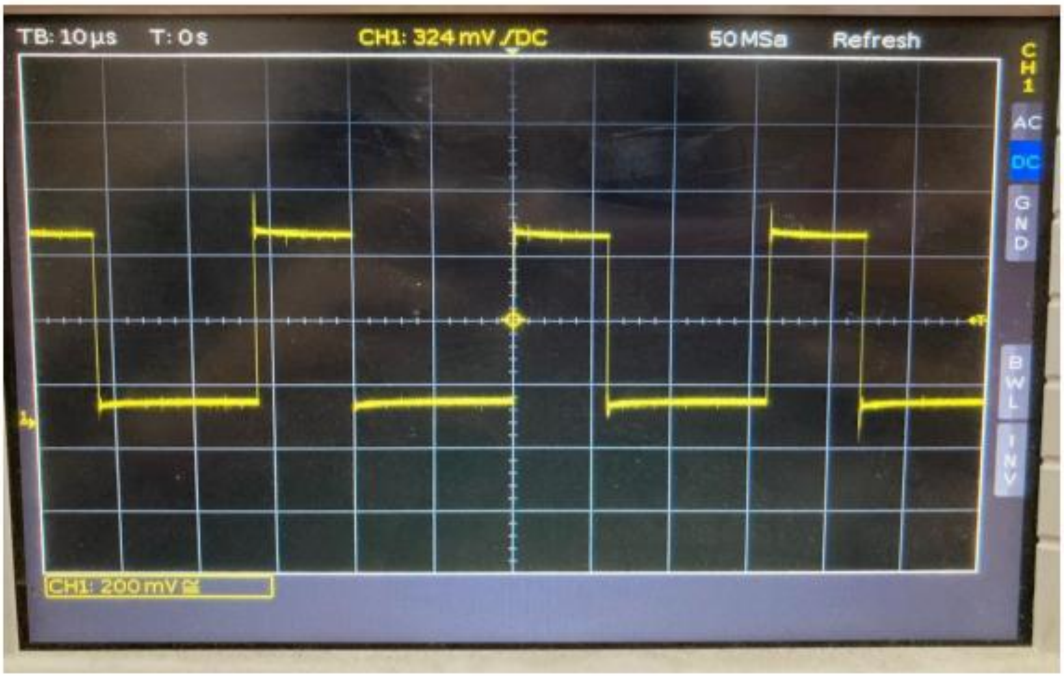


Fig-01: Input Signal

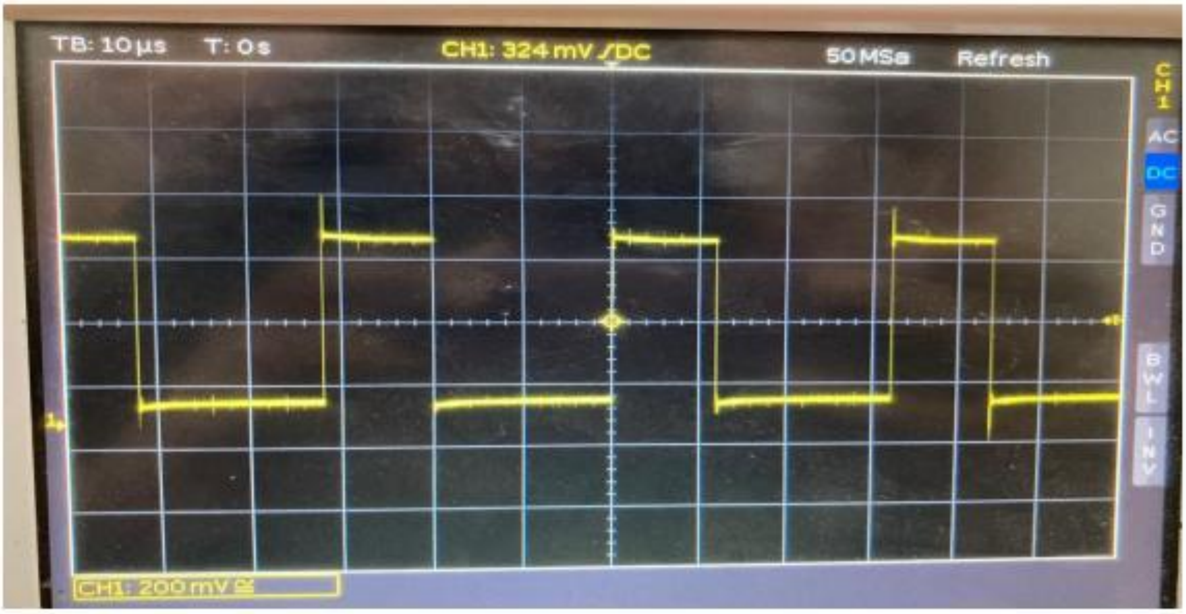


Fig-02: NRZ-L (after encoding)

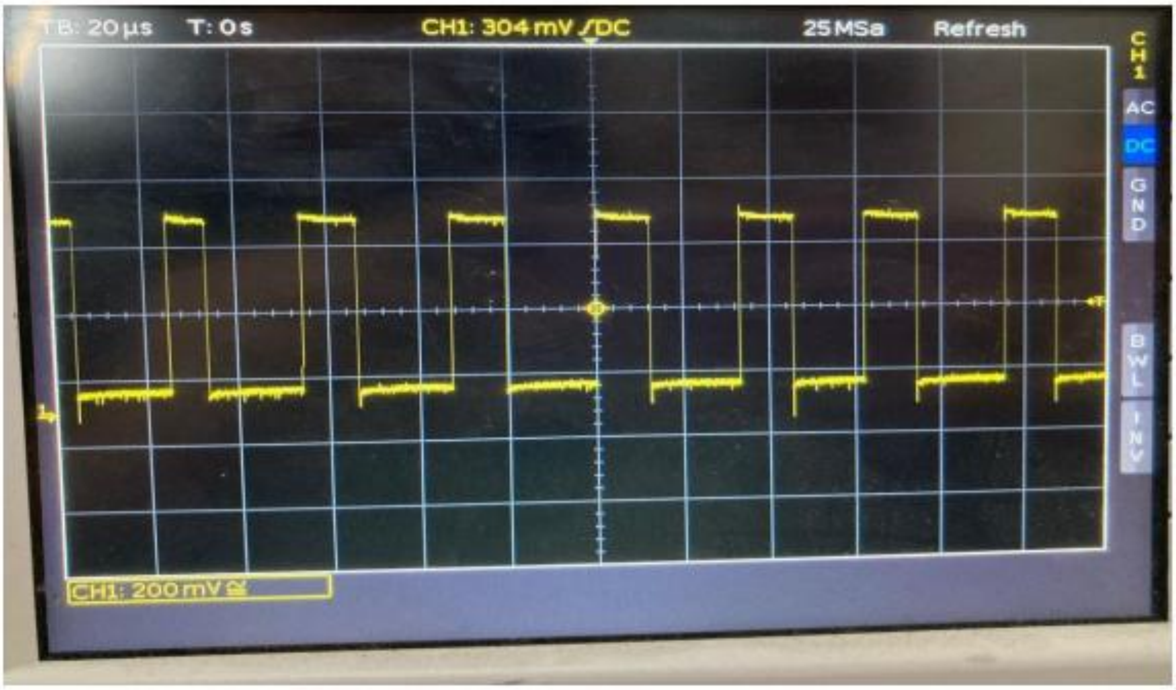


Fig-03: NRZ-L (after decoding)

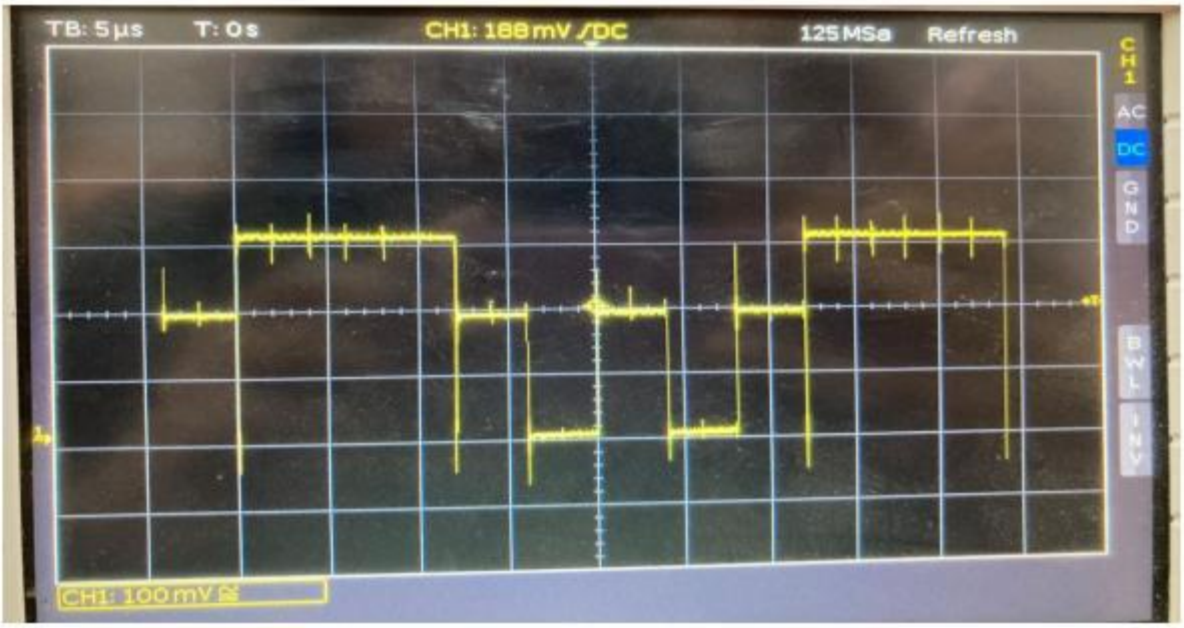


Fig-04: NRZ-S (after encoding)

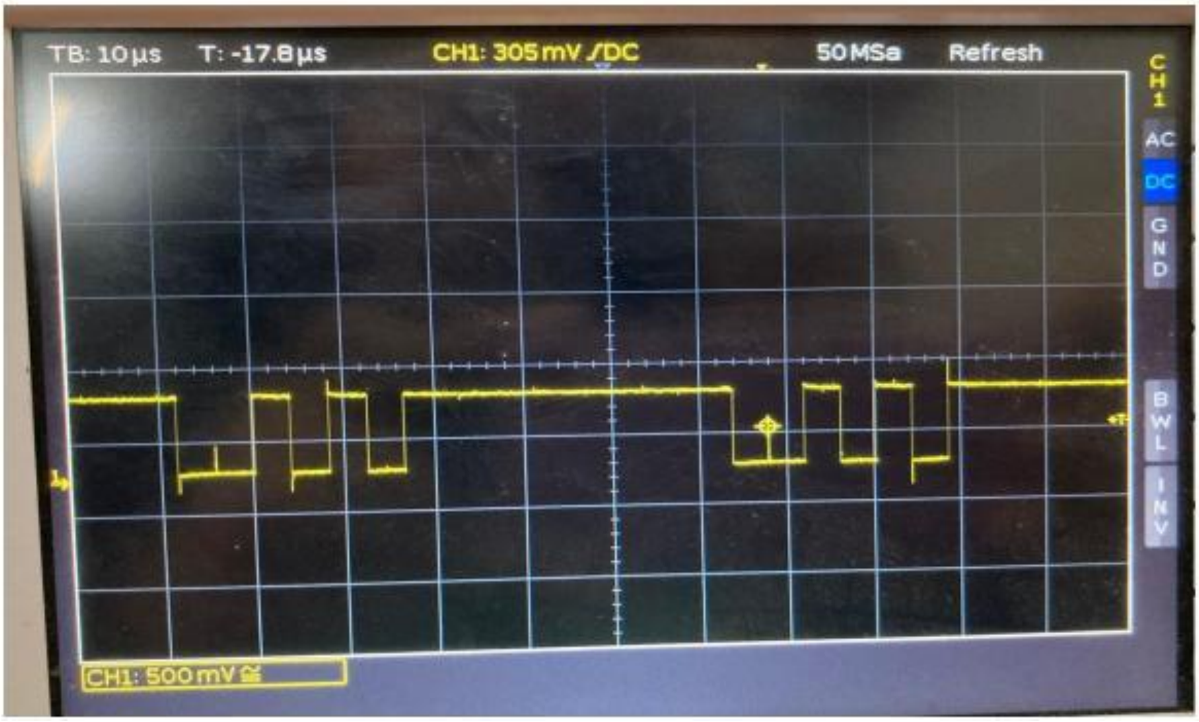


Fig-05: NRZ-S (after decoding)

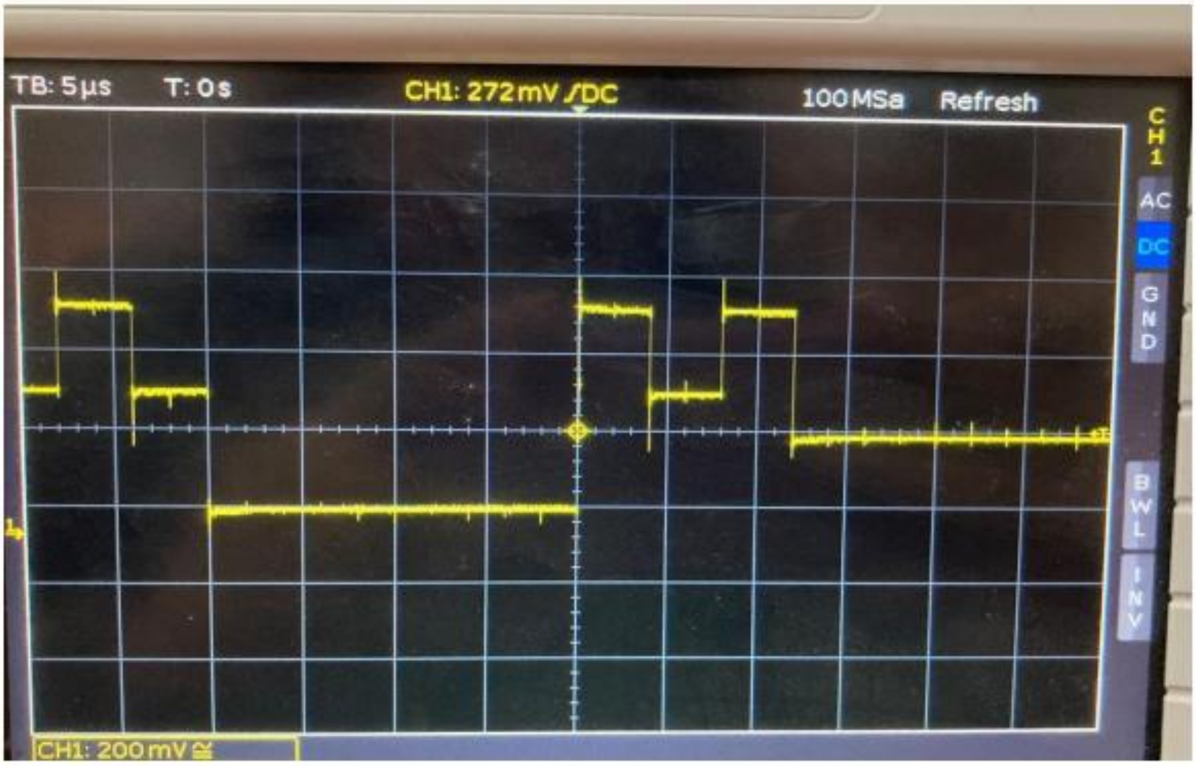


Fig-06: NRZ-M (after encoding)

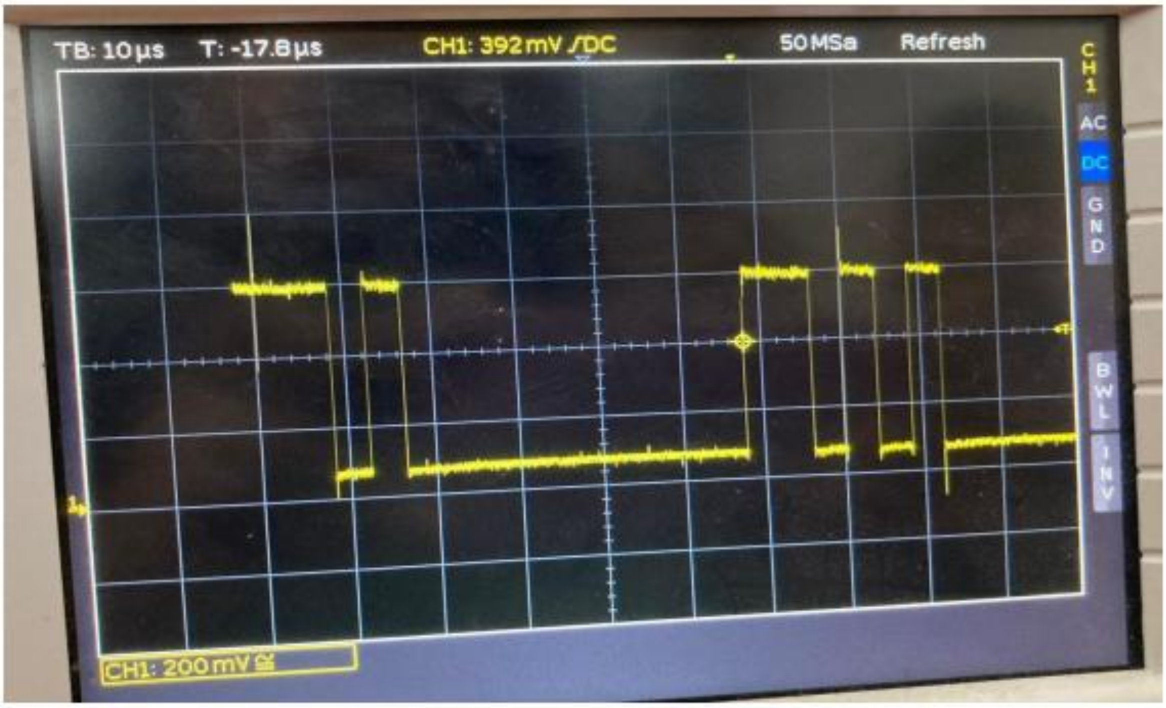


Fig-07: NRZ-M (after decoding)

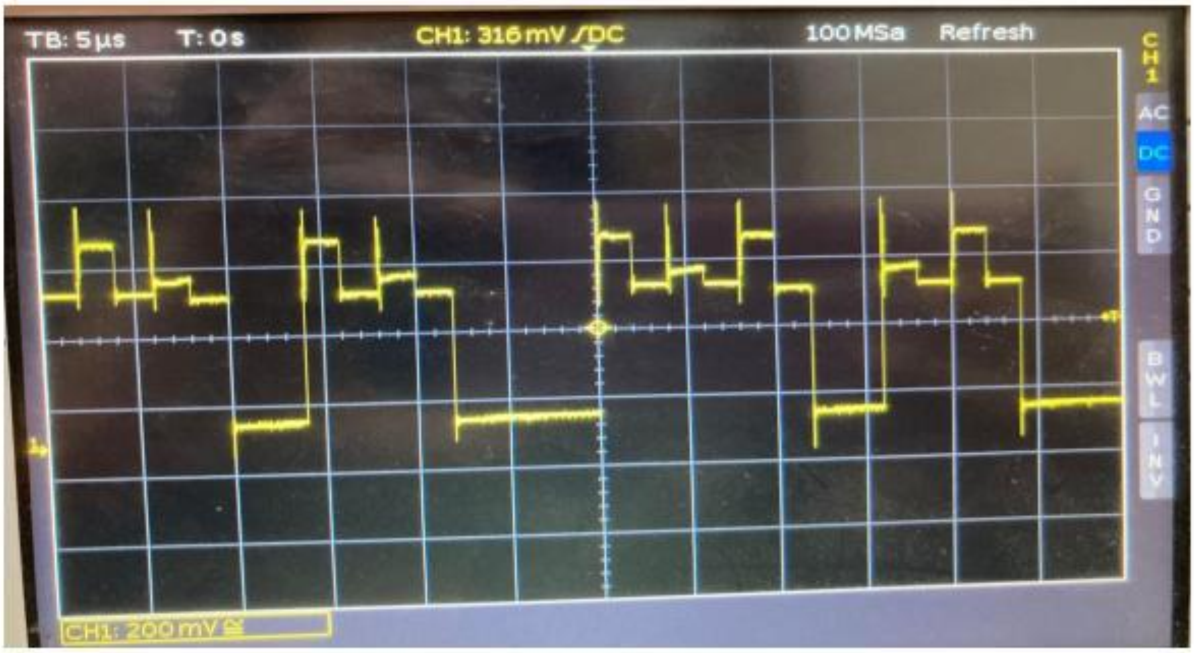


Fig-08: URZ-AMI (after encoding)

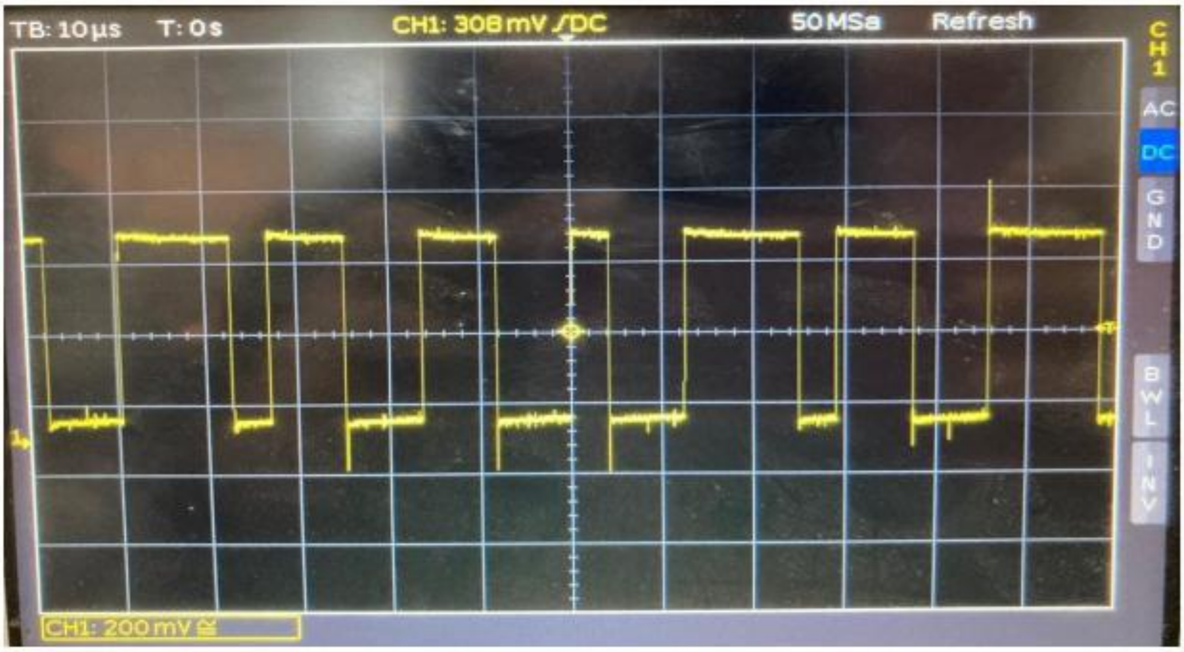


Fig-08: URZ-AMI (after decoding)

Analysis:

- NRZ-L: Clear differentiation between '1' and '0' but lacks synchronization.
- NRZ-S: Effective in reducing the DC component but requires accurate clocking. NRZ-M: Like NRZ-S but relies on transitions for '1', reducing error probability.
- URZ-AMI: Ensures no DC component and maintains synchronization due to alternating polarity for '1'.

Oscilloscope waveforms for each scheme were captured and analyzed for variations in voltage levels and transitions.

Precautions:

- Double-check all circuit connections for correctness.
- Set Frequencies Correctly.
- Properly adjusted the probes of the oscilloscope.
- Handled integrated circuits (ICs) and patch cords carefully to avoid bending pins or breaking wires.

Conclusion:

In conclusion, this experiment successfully validated the encoding and decoding mechanisms of various line coding schemes. The analysis highlighted the specific trade-offs inherent in each method; specifically, URZ-AMI emerged as the superior choice for mitigating DC bias and facilitating clock recovery. Conversely, while NRZ-L proved to be the most architecturally simple, it demonstrated significant limitations regarding synchronization. These findings reinforce the critical importance of selecting the appropriate line coding technique for specific digital communication environments.